

DDTA research work plan after documentation-layer closure

WORK PLAN - REVISION 13

Status: active BA3 execution plan after BA3-T1 provenance/derivation lower-bound pressure testing.

Supersedes: Revision 12 only for forward execution state; R1-R12 remain historical research records.

1. Fixed prior state

- Chapters 2-4: CLOSED / FINAL for current thesis scope.
- Documentation layer: CLOSED.
- BA0-R systems-modeling prior-art research: CLOSED.
- BA0 responsibility/non-goals: CLOSED.
- BA1 minimal BAE identity ontology: CLOSED.
- BA2 relation/action vocabulary: **CLOSED BY BA2-T4.**
- BAreferent and BAProposition: ACCEPTED first-class semantic identity families.
- ThreatForge is a case study/tooling instrument, not DDTA semantic authority.

2. Closed BA2 dependency

BA3 must annotate, trace and govern provenance for the already-closed BA2 proposition semantics. It must not reopen operator/role semantics without a material counterexample.

3. BA3 objective

BA3 defines the smallest derivation/provenance/authority mechanics that connect governed documentation to Base Analysis while preserving:

```
governed documentation = project authority
Base Analysis           = accepted analytical representation
analysis result         = method-owned / reviewable
corrective candidate    = proposed documentation change
```

4. BA3-T1 - source-to-Base-Analysis derivation and provenance lower bound

Status: COMPLETED / PROVISIONAL PASS WITH LOWER-BOUND CANDIDATE.

BA3-T1 pressure-tested provenance target, multiplicity, source locators, baseline context, origin states, derivation basis and authority leakage across the current corpora.

Result

Provenance on BAreferent	REQUIRED
Provenance on BAProposition	REQUIRED
Many-to-many source <-> BA lineage	REQUIRED
Immutable governed-baseline context	REQUIRED
Logical document identity + exact locator	REQUIRED
Source prose copy as BA authority	REJECTED
GROUNDING / DERIVED / DIAGNOSTIC_UNRESOLVED	REQUIRED
Derived basis separate from ultimate authority	REQUIRED
Inspectable derivation rule/rationale reference	REQUIRED CANDIDATE
Exact derivation-rule registry	OPEN

Acceptance/review lifecycle	NOT CLOSED BY T1
New provenance BAE identity family	NOT FORCED
Method finding / ThreatForge state as source	REJECTED
BA1 / BA2 reopen	NOT TRIGGERED

Active lower-bound candidate

```

BAOriginAttachment
|- targetElement          1      BAreferent | BAProposition
|- governedBaselineKey    1
|- originState            1      GROUNDED | DERIVED | DIAGNOSTIC_UNRESOLVED
|- sourceLink             0..*   -> GovernedSourceRef
|- derivationBasis         0..*   -> GovernedSourceRef | BAElementRef
'- derivationRuleRef       0..1   [required for DERIVED; exact form OPEN]

GovernedSourceRef
|- documentIdentity        1
'- locator                 1

```

The contract is representation-independent metadata, not a third BAE family.

5. Why BA3 remains open

T1 closes no lifecycle/equivalence contract. Material unresolved questions remain:

- whether a BA identity is retained or replaced when governed sources change;
- when an existing element becomes stale, superseded or retired;
- how accepted/rejected/review state relates to origin state;
- how a diagnostic is resolved/replaced after source correction;
- exact derivation-rule registry/material form;
- downstream change-impact and feedback lineage.

These cannot be answered by adding more source fields; they require mutation pressure.

6. BA3-T2 - cross-baseline identity, staleness and lifecycle pressure test

NEXT / NOT EXECUTED BY THIS PLAN REVISION.

BA3-T2 must use concrete governed mutations rather than inventing lifecycle vocabulary in the abstract.

It must pressure-test at least:

1. facial M1 Ethernet -> Wi-Fi: retained abstract connectivity meaning versus changed realization proposition/source;
2. facial M2 externally consumed transport -> project-owned transport: responsibility-boundary change and affected proposition identity;
3. facial M3 recognition moved onto camera: retirement/supersession of transfer obligations while unrelated referents may survive;
4. structurally comparable order-fulfillment responsibility-boundary changes where a provider/capability moves internal/external;
5. whether BAreferent and BAProposition use the same retain/revise/replace rules or need distinct lifecycle rules;
6. when an element becomes STALE, SUPERSEDED, RETIRED or remains current after source change;
7. whether per-element ACCEPTED / REJECTED / review state is required separately from origin state;

8. how `DIAGNOSTIC_UNRESOLVED` material is resolved or superseded after governed correction;
9. whether any mutation forces BA1/BA2 reopen rather than only BA3 lifecycle handling.

BA3-T2 exit condition

Produce a falsifiable identity/lifecycle candidate and the smallest unresolved set. Do not close BA3 merely because the mutation examples can be manually explained.

7. BA4 - projections

Only after BA3 closes, define derived human and method projections from the same accepted BA semantics. No view becomes project authority.

8. BA5 - lexical vocabulary and optional assistance

Only after BA3/BA4 boundaries are stable, define display labels, authoring synonyms and optional assistance. NLP/LLM proposals remain candidate-producing support.

9. BA6 - complete Base Analysis regression and closure

Regress the complete BA0-BA5 design against the closed corpora and at least one structurally different holdout. BA6 may reopen only the smallest earlier responsibility on a material counterexample.

10. Analysis envelope remains downstream

Only after Base Analysis closure:

- A1 - AnalysisRecord / execution envelope;
- A2 - common reviewed Finding boundary;
- A3 - change/provenance integration;
- formal methodology overlays and STRIDE/STRIDE-AI evaluations.

The earlier STRIDE transfer probe remains only a bounded consumer probe.

11. Next authorized microstep

Only after the BA3-T1 package is reviewed, committed, pushed and remotely verified, execute:

BA3-T2 - cross-baseline identity, staleness and lifecycle pressure test.

Do not start BA4, formal threat-method overlays, Common Finding schema or implementation work.